

[EECS 304] Control I

Sheet (2) Transfer Function

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Outline

1 Transfer Function

2 Problems

- Problem 1
- Problem 2
- Problem 3
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- Problem 5

Transfer Function

- **Goal:** Develop the mathematical models of physical systems.
- What's meant by mathematical model?
- Why are we concerned with developing models for physical systems?
 - 1 Predict the system response under certain conditions without testing the physical system.
 - 2 Illustrate the analysis and design of control systems.

Transfer Function

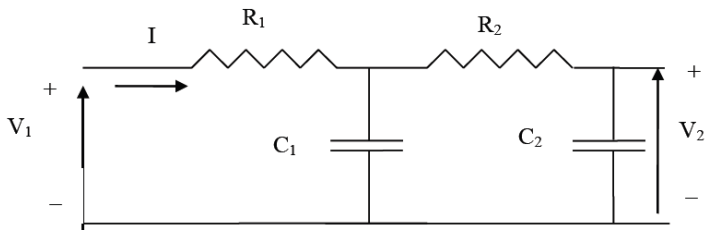
Consider a system with input $e(t)$ and output $c(t)$. If $C(s)$ is the Laplace transform of the system output and $E(s)$ is the Laplace transform of the system input, the transfer function $G(s)$ is defined by

$$G(s) = \frac{C(s)}{E(s)}$$

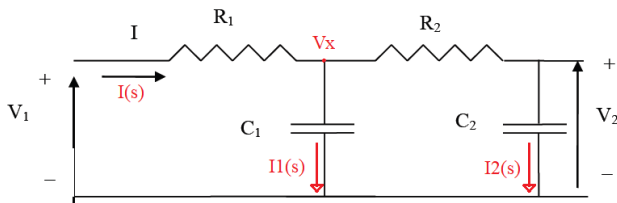
Problem 1

Problem 1

For the circuit shown, obtain the transfer function $V_2(s)/V_1(s)$.



Problem 1



$$\therefore V_1(s) = I(s)R_1 + V_x(s)$$

$$\therefore V_x(s) = I_2(s)R_2 + V_2(s)$$

$$\therefore V_1(s) = I(s)R_1 + I_2(s)R_2 + V_2(s)$$

$$\therefore I(s) = I_1(s) + I_2(s)$$

$$\therefore V_1(s) = [I_1(s) + I_2(s)]R_1 + I_2(s)R_2 + V_2(s)$$

$$\therefore I_1(s) = V_x(s)sC_1 \quad \therefore I_2(s) = V_2(s)sC_2$$

$$\therefore V_1(s) = [V_x(s)sC_1 + V_2(s)sC_2]R_1 + [V_2(s)sC_2]R_2 + V_2(s)$$

Problem 1

$$\therefore V_1(s) = V_x(s)sC_1R_1 + V_2(s)sC_2R_1 + V_2(s)sC_2R_2 + V_2(s)$$

$$\therefore V_1(s) = [I_2(s)R_2 + V_2(s)]sC_1R_1 + V_2(s)sC_2R_1 + V_2(s)sC_2R_2 + V_2(s)$$

$$\therefore V_1(s) = I_2(s)R_2sC_1R_1 + V_2(s)sC_1R_1 + V_2(s)sC_2R_1 + V_2(s)sC_2R_2 + V_2(s)$$

$$\therefore V_1(s) = [V_2(s)sC_2]R_2sC_1R_1 + V_2(s)sC_1R_1 + V_2(s)sC_2R_1 + V_2(s)sC_2R_2 + V_2(s)$$

$$\therefore V_1(s) = V_2(s)s^2C_2R_2C_1R_1 + V_2(s)sC_1R_1 + V_2(s)sC_2R_1 + V_2(s)sC_2R_2 + V_2(s)$$

$$\therefore V_1(s) = V_2(s)[s^2R_1R_2C_1C_2 + sR_1C_1 + sR_1C_2 + sR_2C_2 + 1]$$

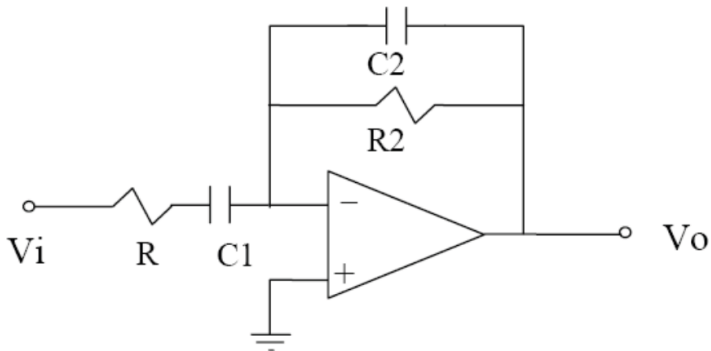
Answer:

$$\therefore \frac{V_2(s)}{V_1(s)} = \frac{1}{s^2R_1R_2C_1C_2 + sR_1C_1 + sR_1C_2 + sR_2C_2 + 1}$$

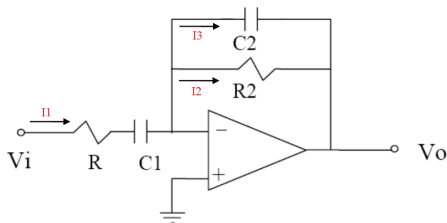
Problem 2

Problem 2

For the ideal OP Amp, obtain the transfer function $V_o(s)/V_i(s)$.



Problem 2



$$\therefore I_1(s) = I_2(s) + I_3(s)$$

$$\therefore I_1(s) = \frac{V_{in}(s)}{R_1 + \frac{1}{sC_1}}$$

$$\therefore I_2(s) = \frac{0 - V_o(s)}{R_2} \quad \therefore I_3(s) = \frac{0 - V_o(s)}{\frac{1}{sC_2}}$$

Problem 2

$$\therefore \frac{V_{in}(s)}{R_1 + \frac{1}{sC_1}} = \frac{-V_o(s)}{R_2} - V_o(s)sC_2$$

$$\therefore \frac{V_o(s)}{V_{in}(s)} = \frac{-1}{(R_1 + \frac{1}{sC_1})(\frac{1}{R_2} + sC_2)}$$

$$\therefore \frac{V_o(s)}{V_{in}(s)} = \frac{-1}{sC_2R_1 + \frac{1}{sC_1R_2} + \frac{C_2}{C_1} + \frac{R_1}{R_2}}$$

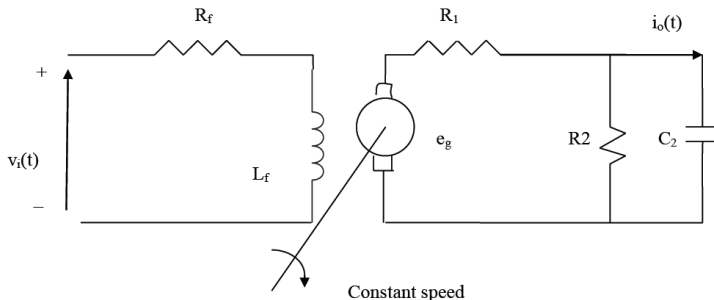
Answer:

$$\therefore \frac{V_o(s)}{V_{in}(s)} = \frac{-sC_1R_2}{s^2C_1C_2R_1R_2 + sC_2R_2 + sC_1R_1 + 1}$$

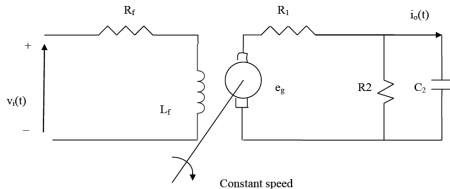
Problem 3

Problem 3

The system shown consists of a DC generator connected to an RC impedance. Find the TF that relates V_i and I_o .



Problem 3



- The current i_f flowing through the coil will produce a magnetic flux $\phi \rightarrow \phi \propto i_f$
- The generator plate rotating with an angular speed $\dot{\theta}$ through the flux ϕ will generate emf that depends on $\dot{\theta}$ and ϕ

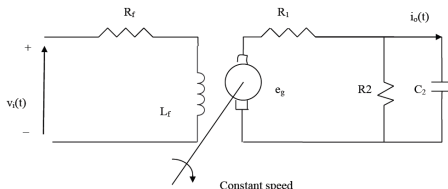
$$e_g \propto \dot{\theta} \phi \rightarrow \because \phi \propto i_f$$

$$\therefore e_g = k \dot{\theta} i_f$$

$\therefore$ the plate rotates with a constant speed

$$\therefore e_g = K i_f \text{ where } K \text{ is constant}$$

Problem 3



$$v_i(t) = i_f(t)R_f + L_f \frac{di_f}{dt}$$

$$e_g(t) = K i_f(t)$$

$$e_g(t) = R_1[i_o(t) + \frac{v_1(t)}{R_2}] + v_1(t)$$

$$i_o(t) = C_2 \frac{dv_1(t)}{dt}$$

$$V_i(s) = (R_f + L_f s) I_f(s)$$

$$E_g(s) = K I_f(s)$$

$$E_g(s) = R_1[I_o(s) + \frac{V_1(s)}{R_2}] + V_1(s)$$

$$I_o(s) = C_2 s V_1(s)$$

Problem 3

$$\therefore V_i(s) = (R_f + L_f s)I_f(s) \rightarrow I_f(s) = V_i(s) \frac{1}{R_f + L_f s}$$

$$\therefore E_g(s) = K I_f(s) \rightarrow E_g(s) = V_i(s) \frac{K}{R_f + L_f s}$$

$$\therefore E_g(s) = V_1(s) \left[\frac{R_1}{R_2} + 1 \right] + R_1 I_o(s)$$

$$\therefore I_o(s) = C_2 s V_1(s) \rightarrow V_1(s) = I_o(s) \frac{1}{s C_2}$$

$$\therefore E_g(s) = I_o(s) \frac{1}{s C_2} \left[\frac{R_1}{R_2} + 1 \right] + R_1 I_o(s)$$

$$\therefore I_o(s) = E_g(s) \frac{1}{R_1 + \frac{1}{s C_2} \left[\frac{R_1}{R_2} + 1 \right]}$$

Problem 3

$$\therefore I_o(s) = V_i(s) \frac{K}{(R_f + sL_f)(R_1 + \frac{1}{sC_2}[\frac{R_1}{R_2} + 1])}$$

$$\therefore \frac{I_o(s)}{V_i(s)} = \frac{K}{(R_f + sL_f)(R_1 + \frac{1}{sC_2}[\frac{R_1}{R_2} + 1])}$$

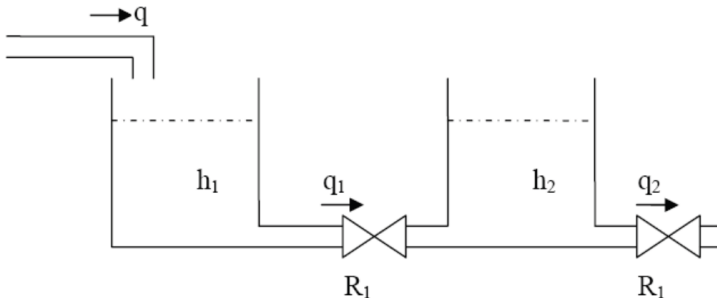
Answer:

$$\therefore \frac{I_o(s)}{V_i(s)} = \frac{sC_2K}{(R_f + sL_f)(sC_2R_1 + \frac{R_1}{R_2} + 1)}$$

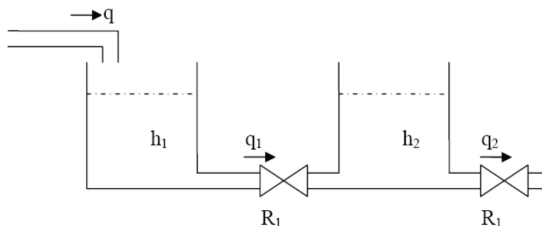
Problem 4

Problem 4

The figure shows a process plant containing of two tanks of areas A_1 and A_2 . Derive the transfer function that relates q_2 to q .

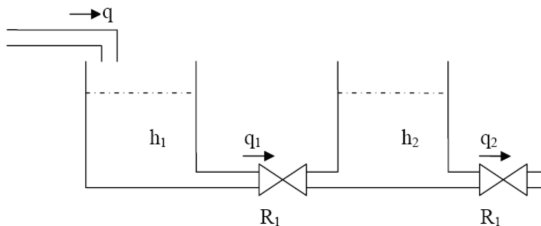


Problem 4



- q_{in} : flow rate input to tank (m^3/sec)
- q_{out} : flow rate output from tank (m^3/sec)
- A : cross section area of the tank (m^2)
- h : liquid level (m)
- h_{next} : liquid level of the next tank (m)
- R_{in} : resistance of the valve controlling the input liquid (ohm)

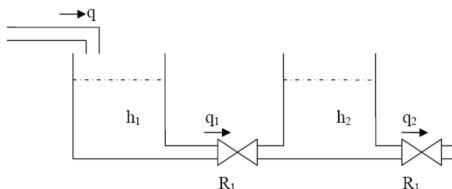
Problem 4



For each tank, we should apply:

- ① $q_{in}(t) - q_{out}(t) = A \frac{dh}{dt}$
- ② $\frac{h(t) - h_{next}(t)}{R_{out}} = q_{out}(t)$

Problem 4



$$q - q_1 = A_1 \frac{dh_1}{dt}$$

$$q_1 - q_2 = A_2 \frac{dh_2}{dt}$$

$$\frac{h_1(t) - h_2(t)}{R_1} = q_1(t)$$

$$\frac{h_2(t) - 0}{R_1} = q_2(t)$$

$$Q(s) - Q_1(s) = A_1 s H_1(s)$$

$$Q_1(s) - Q_2(s) = A_2 s H_2(s)$$

$$\frac{H_1(s) - H_2(s)}{R_1} = Q_1(s)$$

$$H_2(s) = R_1 Q_2(s)$$

Problem 4

$$\therefore Q(s) - Q_1(s) = A_1 s H_1(s) \rightarrow Q_1(s) = Q(s) - A_1 s H_1(s)$$

$$\therefore Q_1(s) - Q_2(s) = A_2 s H_2(s) \rightarrow Q_2(s) = Q_1(s) - A_2 s H_2(s)$$

$$\therefore Q_2(s) = Q(s) - A_1 s H_1(s) - A_2 s H_2(s)$$

$$Q_2(s) = Q(s) - A_1 s [Q_1(s) R_1 + H_2(s)] - A_2 s [R_1 Q_2(s)]$$

$$Q_2(s) = Q(s) - A_1 s [[Q_2(s) + A_2 s H_2(s)] R_1 + R_1 Q_2(s)] - A_2 s R_1 Q_2(s)$$

$$Q_2(s) = Q(s) - A_1 s Q_2(s) R_1 - A_1 A_2 s^2 H_2(s) R_1 - A_1 s R_1 Q_2(s) - A_2 s R_1 Q_2(s)$$

$$Q_2(s) = Q(s) - A_1 s Q_2(s) R_1 - A_1 A_2 s^2 [R_1 Q_2(s)] R_1 - A_1 s R_1 Q_2(s) - A_2 s R_1 Q_2(s)$$

$$Q_2(s) [1 + A_1 s R_1 + A_1 A_2 s^2 R_1^2 + A_1 s R_1 + A_2 s R_1] = Q(s)$$

Answer:

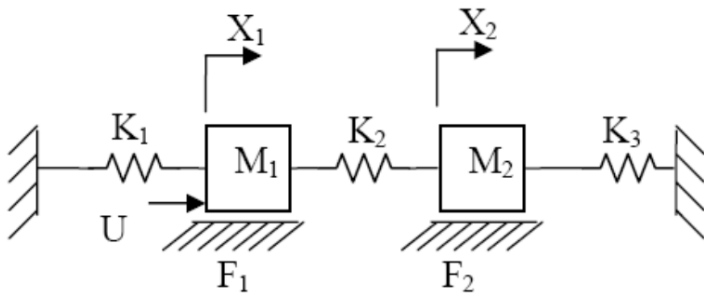
$$\therefore \frac{Q_2(s)}{Q(s)} = \frac{1}{A_1 A_2 R_1^2 s^2 + (2A_1 R_1 + A_2 R_1)s + 1}$$

Problem 5

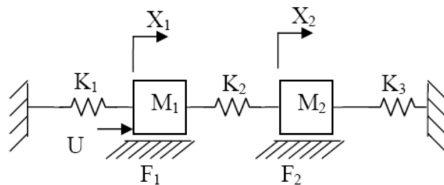
Problem 5

Derive the differential equations that represent the mechanical system shown, where U is a force that affects the mass M_1 and hence derive the transfer functions $X_1(s)/U(s)$ and $X_2(s)/U(s)$.

If the force U has a value of 1 N find the steady state values of X_1 and X_2 .



Problem 5



- K : spring constant (N/m or Kg/s)
- F or β : viscous function coefficient ($N/m/s$ or Kg/s)
- X : displacement due to U or $f(t) \rightarrow$ Tension Force

The mass will oscillate under an acceleration $a = \ddot{x}$, so the equilibrium modelling equation of the system is

$$f(t) - F_1\dot{x} - Kx = M\ddot{x}$$

$$\sum \text{Force} = Ma = M\ddot{x}$$

Problem 5

Applying $\sum \text{Force} = M\ddot{x}$ at each M:

At M_1 :

$$u - K_1x_1 - F_1\dot{x}_1 - K_2(x_1 - x_2) = M_1\ddot{x}_1$$

$$\therefore M_1\ddot{x}_1 + F_1\dot{x}_1 + K_1x_1 + K_2(x_1 - x_2) - u = 0$$

At M_2 :

$$-K_2(x_2 - x_1) - K_3x_2 - F_2\dot{x}_2 = M_2\ddot{x}_2$$

$$\therefore M_2\ddot{x}_2 + F_2\dot{x}_2 + K_3x_2 + K_2(x_2 - x_1) = 0$$

Taking Laplace Transform for both equations:

$$\therefore X_1(s)[M_1s^2 + F_1s + K_1 + K_2] = U(s) + K_2X_2(s)$$

$$\therefore X_2(s)[M_2s^2 + F_2s + K_3 + K_2] = K_2X_1(s)$$

Problem 5

$$\therefore X_1(s)[M_1s^2 + F_1s + K_1 + K_2] = U(s) + K_2\left[\frac{K_2X_1(s)}{M_2s^2 + F_2s + K_3 + K_2}\right]$$

$$\therefore X_1(s)\left[M_1s^2 + F_1s + K_1 + K_2 - \frac{K_2^2}{M_2s^2 + F_2s + K_3 + K_2}\right] = U(s)$$

$$\therefore \frac{X_1(s)}{U(s)} = \frac{M_2s^2 + F_2s + K_3 + K_2}{(M_1s^2 + F_1s + K_1 + K_2)(M_2s^2 + F_2s + K_3 + K_2) - K_2^2}$$

Note:

Steady state of $X_1 \rightarrow \lim_{s \rightarrow 0} sX_1(s)$

$$\therefore X_1|_{ss} = \frac{K_3 + K_2}{(K_1 + K_2)(K_3 + K_2) - K_2^2}$$

Problem 5

$$\therefore \frac{X_2(s)}{U(s)} = \frac{K_2}{(M_1 s^2 + F_1 s + K_1 + K_2)(M_2 s^2 + F_2 s + K_3 + K_2) - K_2^2}$$

$$\therefore X_2|_{ss} = \frac{K_2}{(K_1 + K_2)(K_3 + K_2) - K_2^2}$$

Thank you!

Questions?

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